

Computazione e Fisica

ovvero:

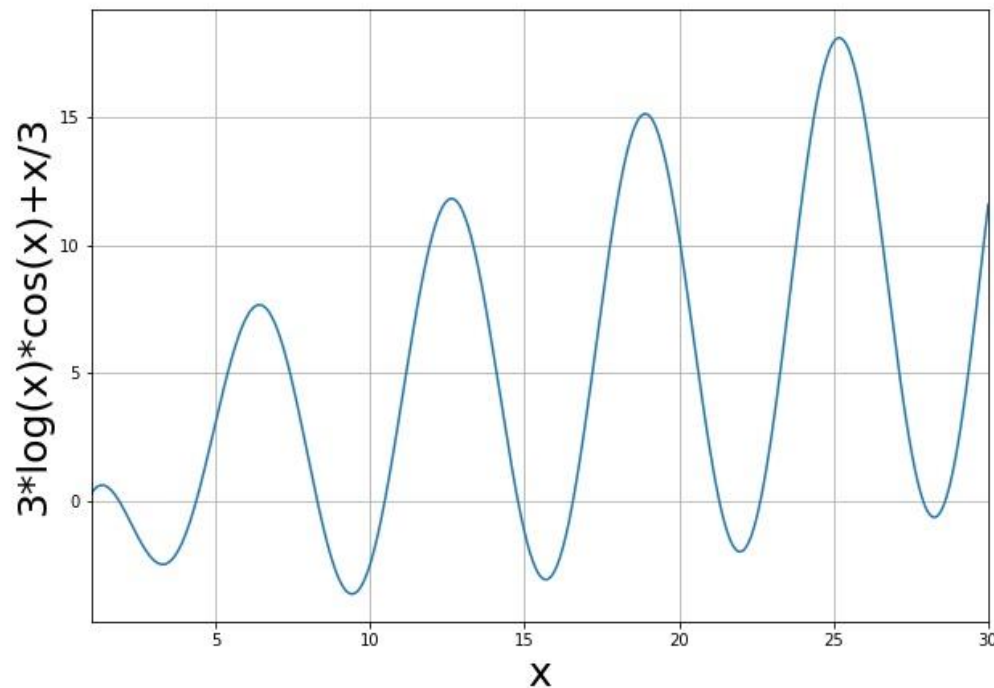
perchè Tecniche Informatiche
al primo semestre del primo anno?

<http://www.to.infn.it/~maina/> → AA 2026/27:TIF

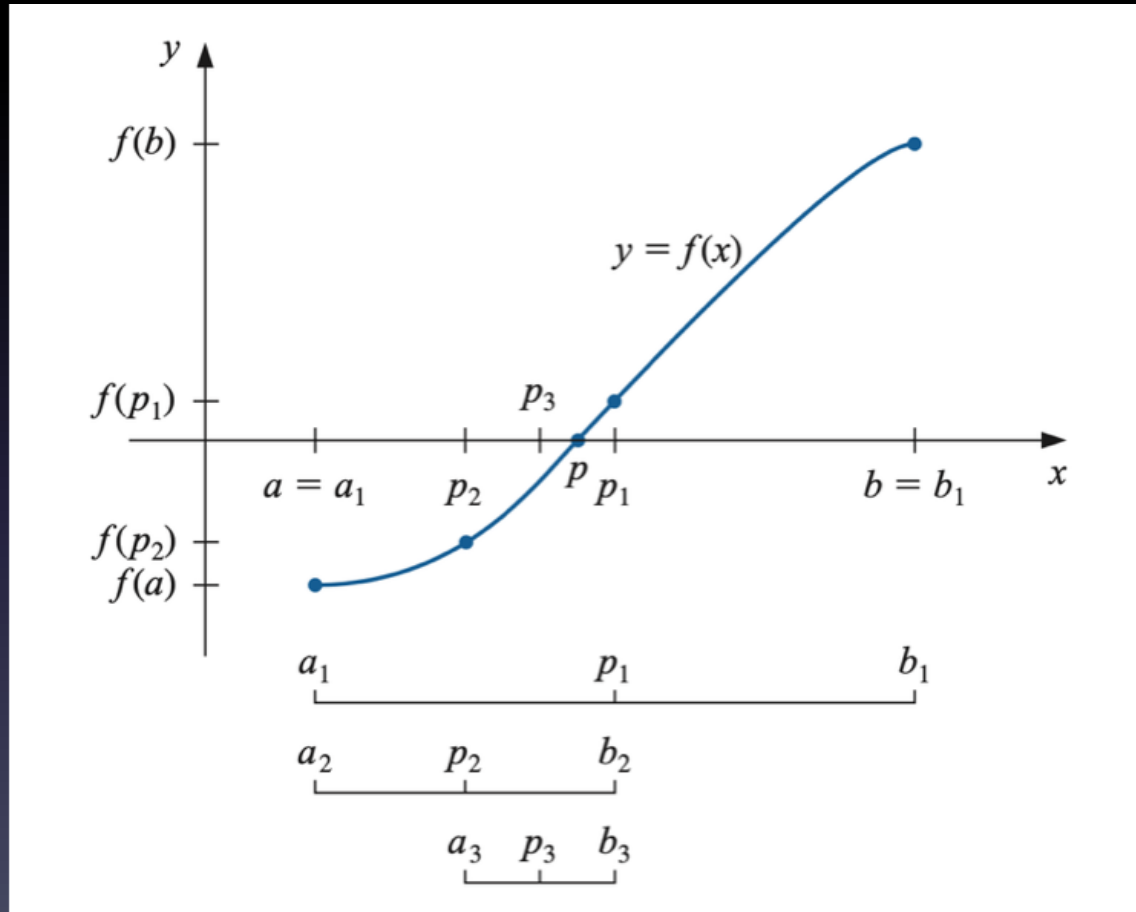
Problemi semplici non risolvibili analiticamente

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0$$

$$n > 4$$

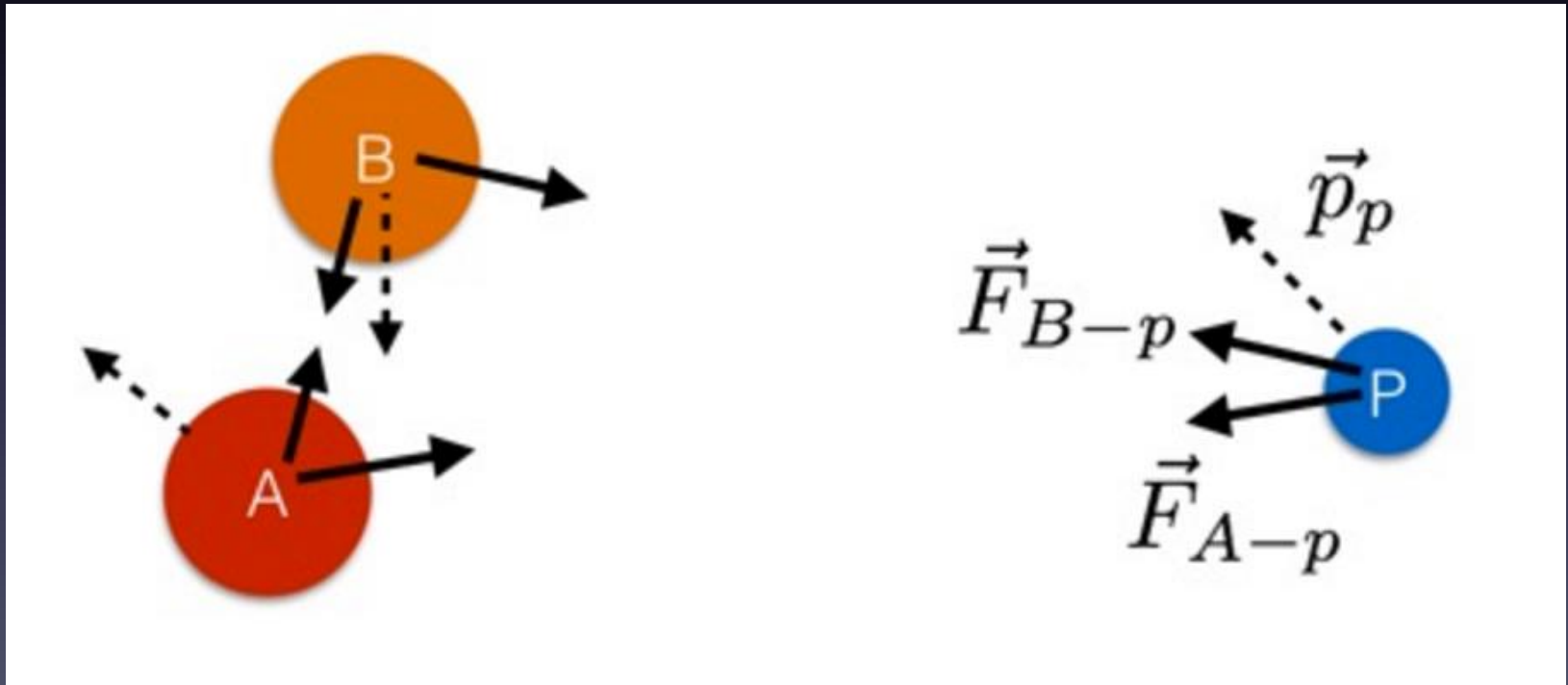


Trovare numericamente una radice di una funzione



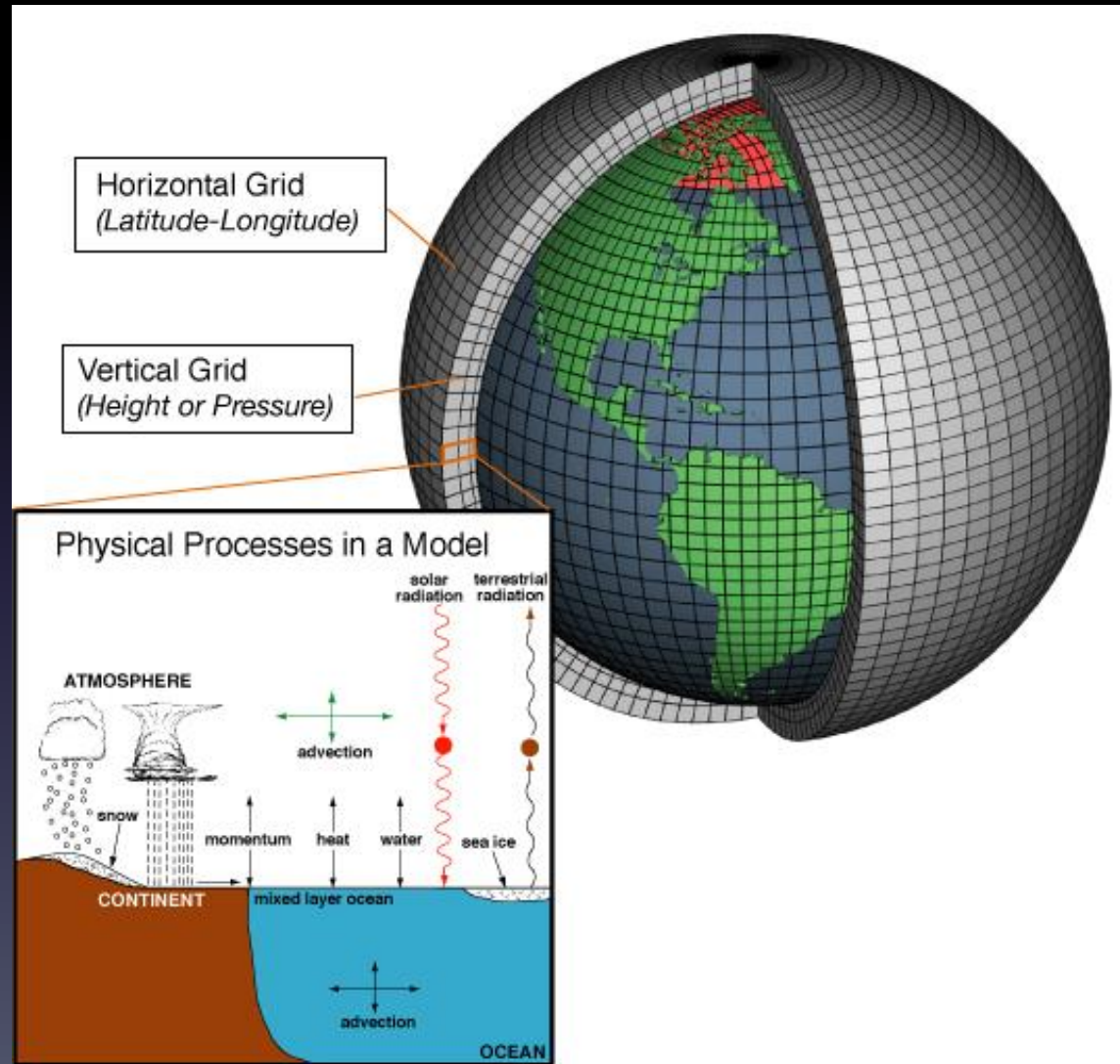
3 masse gravitanti: le forze sono note.

Non sappiamo descrivere analiticamente
il moto.



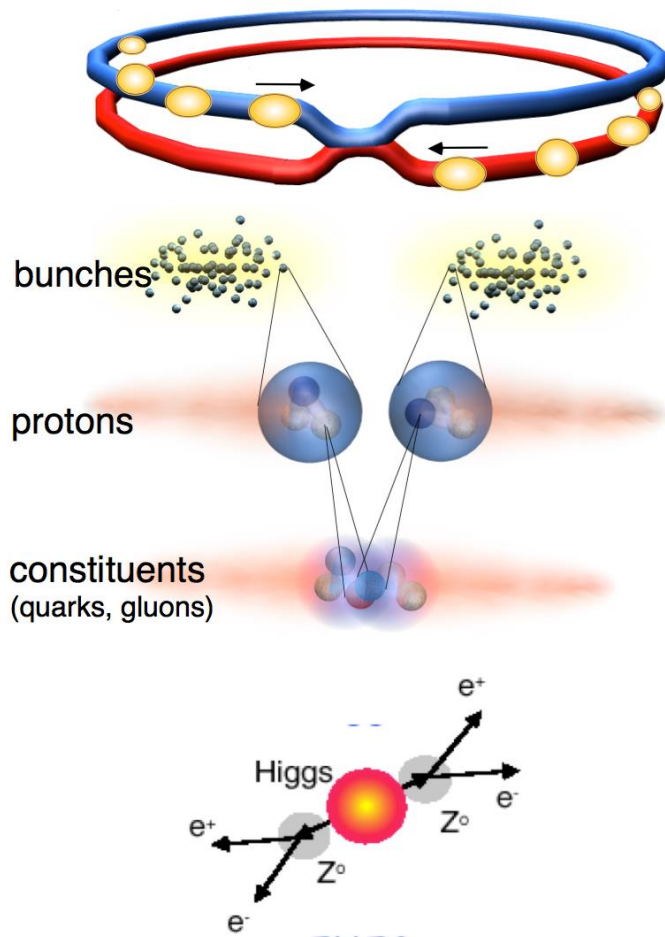
Problemi complessi

Dinamica dell'atmosfera



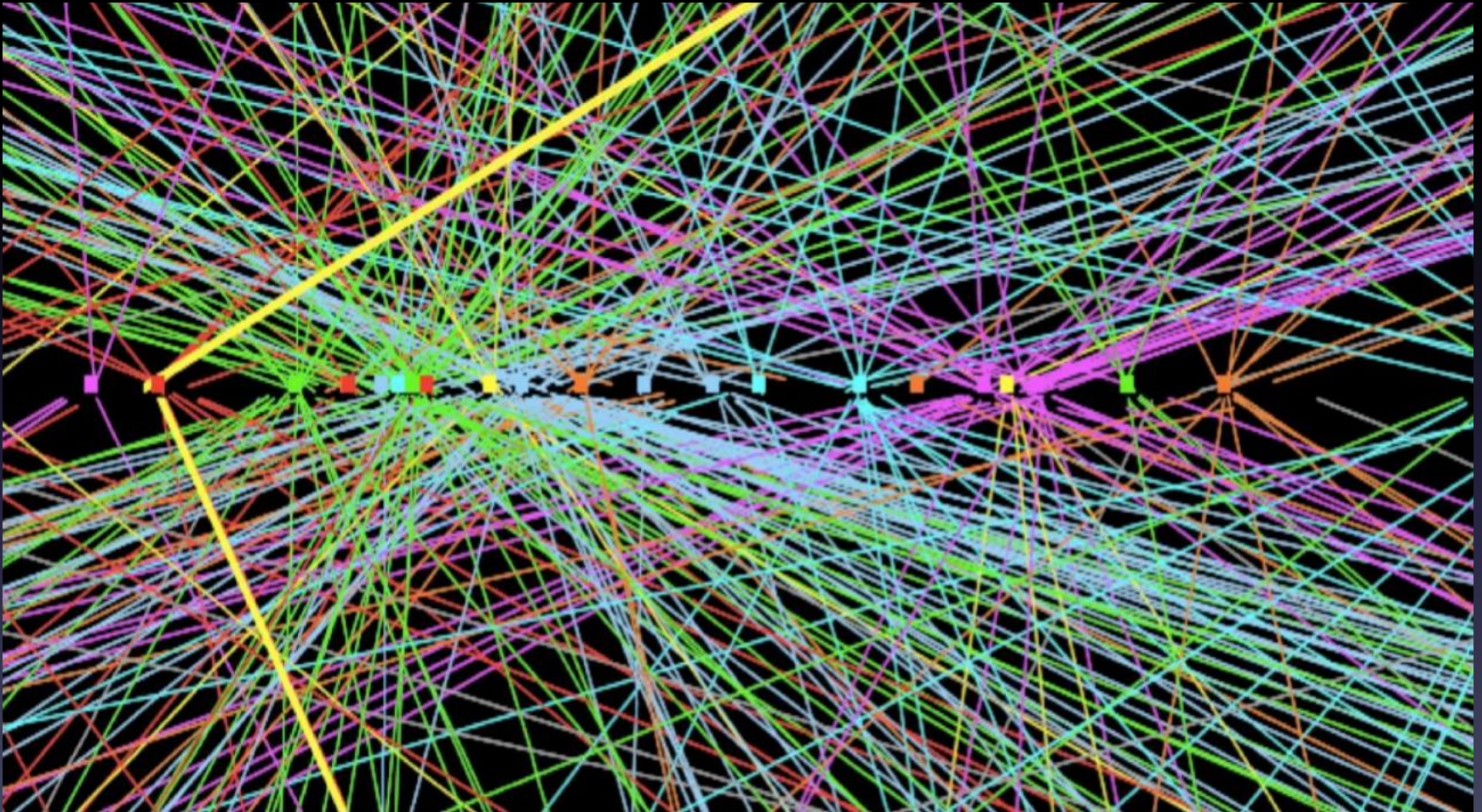
Un esperimento al CERN

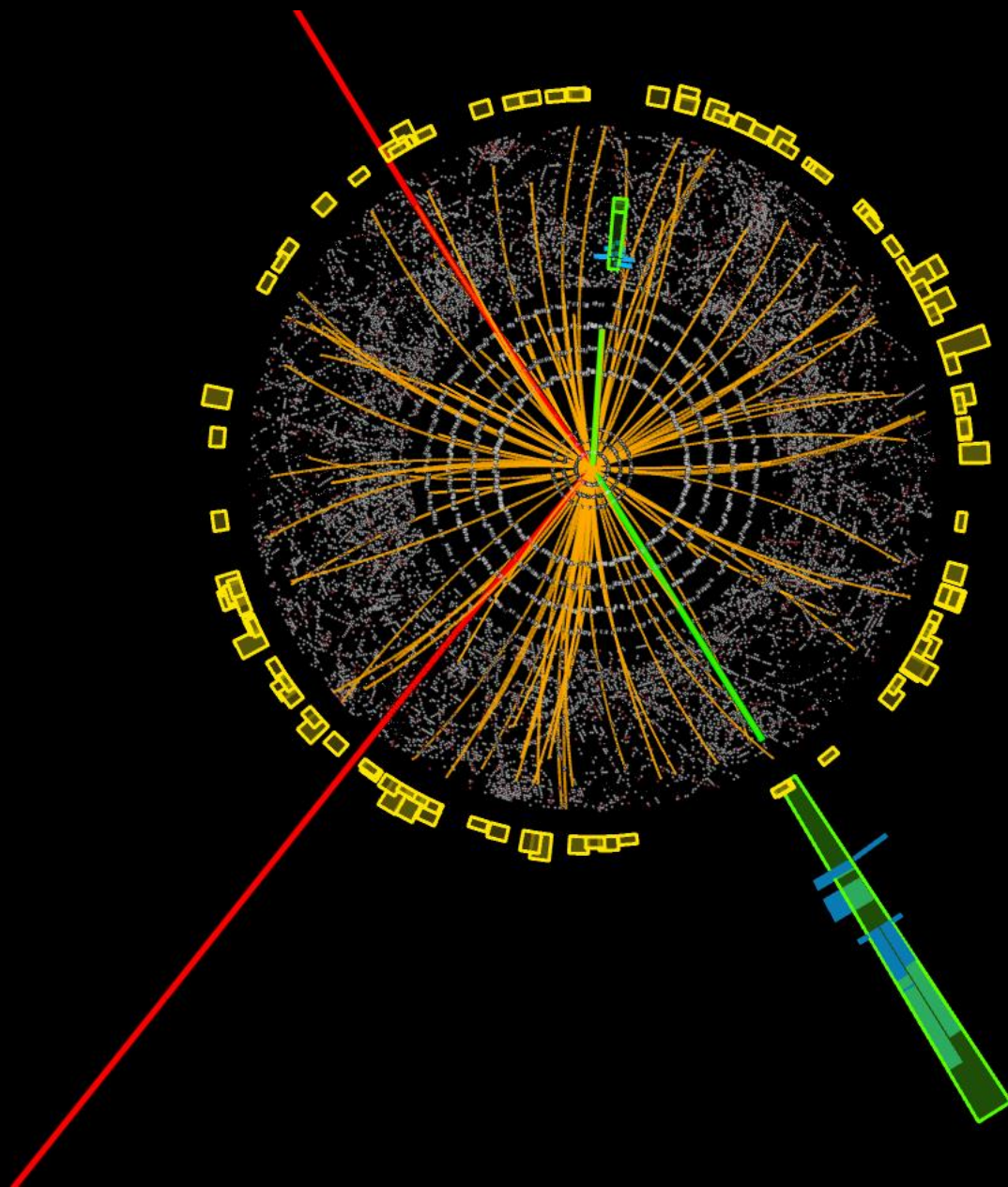
proton collisions at LHC



- 2800 bunches of protons
- energy of each proton : 6.5 TeV
- 100 billions protons / bunch
- beam crossing rate: 40 MHz
- in the experiments at each crossing:
 - ~ 20-50 proton-proton collisions
 - ~ 1500 particles produced
- 1 billion interactions / second
- impossible to record everything !
- a Higgs boson to find within 5 billions of collisions...

Un evento: ~ 20 urti, ogni traccia una particella

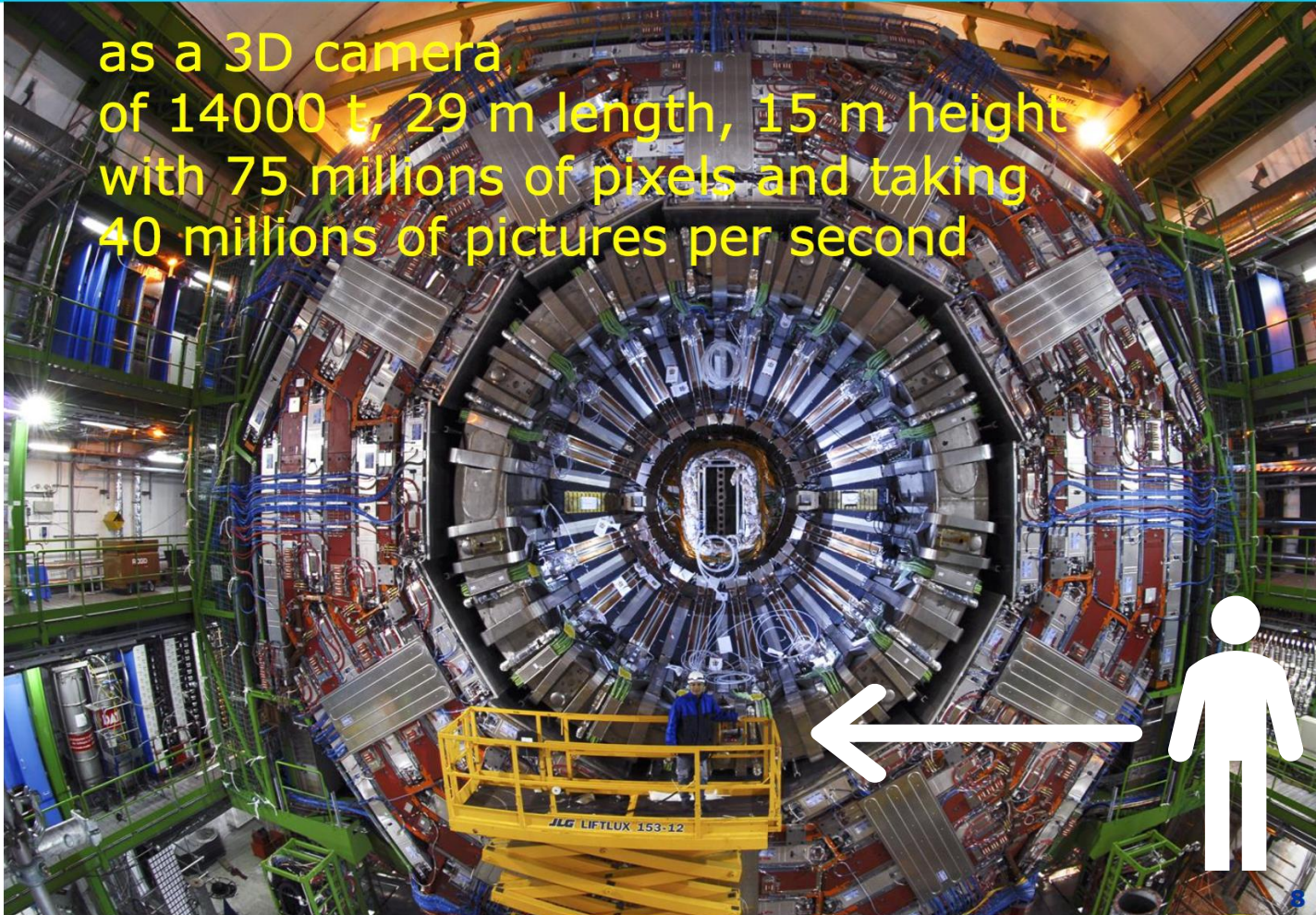




Run: 205113
Event: 12611816
Date: 2012-06-18
Time: 11:07:47 CEST

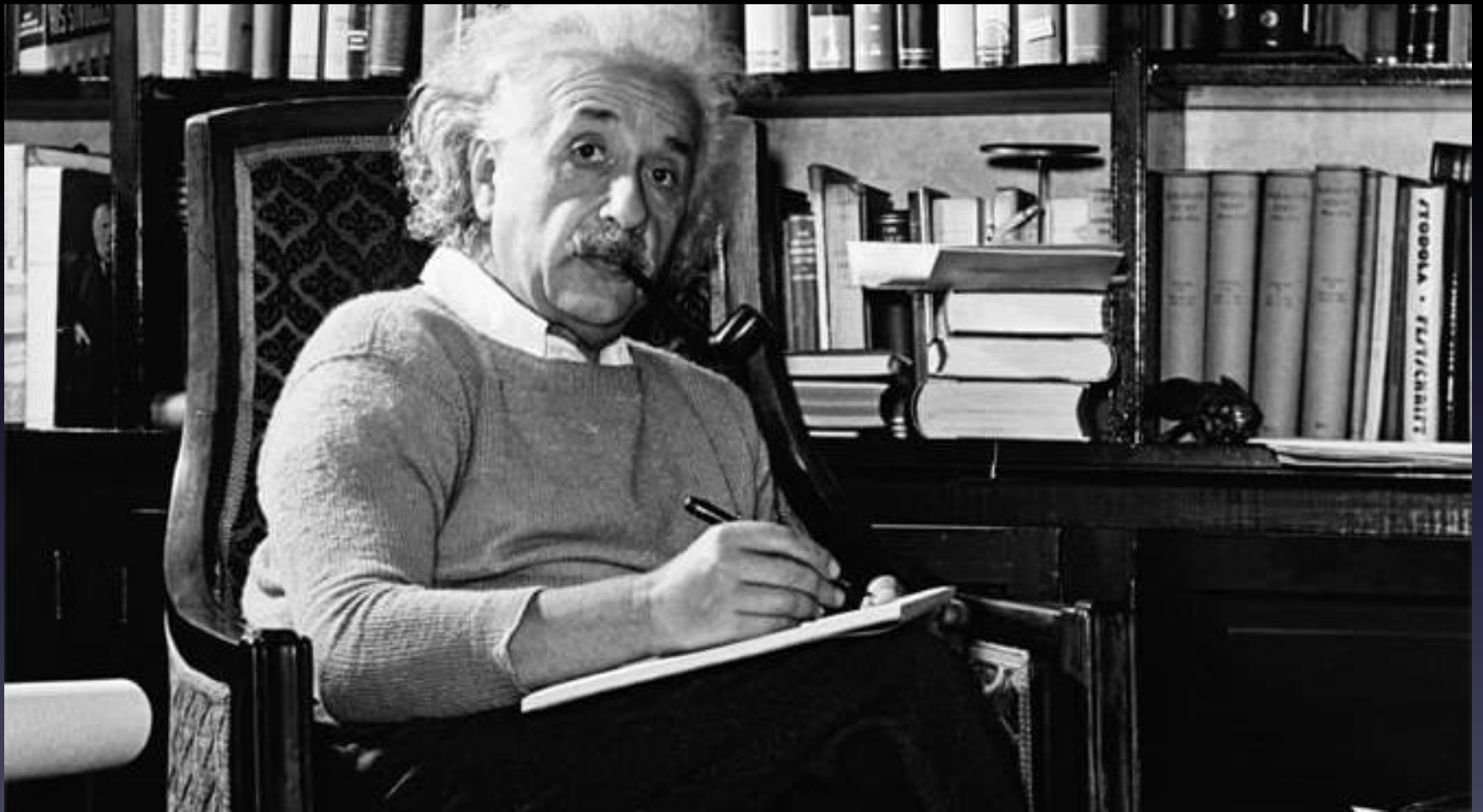
the CMS detector

as a 3D camera
of 14000 t, 29 m length, 15 m height
with 75 millions of pixels and taking
40 millions of pictures per second



Calcolatori indispensabili per trigger, presa dati, analisi dati, simulazioni

Farò il teorico, come Einstein...



Two-Loop Master Integrals for $\gamma^* \rightarrow 3$ Jets: The planar topologies

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Agosto 2000



$$= \left(\frac{S_\epsilon}{16\pi^2} \right)^2 \frac{(-s_{123})^{-2\epsilon}}{s_{12} + s_{13}} \sum_{i=-1}^3 \frac{f_{5.1,i} \left(\frac{s_{13}}{s_{123}}, \frac{s_{23}}{s_{123}} \right)}{\epsilon^i} + \mathcal{O}(\epsilon^2), \quad (4.20)$$

with:

$$f_{5.1,3}(y, z) = 0, \quad (4.21)$$

$$f_{5.1,2}(y, z) = -H(0; z), \quad (4.22)$$

$$f_{5.1,1}(y, z) = +H(0; y)H(0; z) - 2H(0; z) + 2H(0, 0; z) + H(1, 0; z) + \frac{\pi^2}{6}, \quad (4.23)$$

$$\begin{aligned} f_{5.1,0}(y, z) = & +2H(0; y)H(0; z) - 2H(0; y)H(1, 0; z) - 4H(0; z) - H(0; z)H(1 - z, 0; y) \\ & - 2H(0, 0; y)H(0; z) + 4H(0, 0; z) - 2H(0, 0; z)H(0; y) - 4H(0, 0, 0; z) - H(0, 1, 0; y) \\ & - 2H(0, 1, 0; z) + 2H(1, 0; z) - H(1, 0; z)H(1 - z; y) - 2H(1, 0, 0; z) - 2H(1, 1, 0; z) \\ & - H(1 - z, 1, 0; y) + \frac{\pi^2}{6} [+2 - 2H(0; y) - 3H(0; z) - 2H(1; z) - H(1 - z; y)], \end{aligned} \quad (4.24)$$

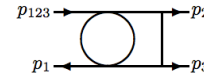
$$\begin{aligned} f_{5.1,-1}(y, z) = & +4H(0; y)H(0; z) - 4H(0; y)H(1, 0; z) + 4H(0; y)H(1, 0, 0; z) + 4H(0; y)H(1, 1, 0; z) \\ & - 8H(0; z) - 2H(0; z)H(1 - z, 0; y) + 2H(0; z)H(1 - z, 0, 0; y) \\ & + H(0; z)H(1 - z, 1 - z, 0; y) - 4H(0, 0; y)H(0; z) + 4H(0, 0; y)H(0, 0; z) \\ & + 4H(0, 0; y)H(1, 0; z) + 8H(0, 0; z) - 4H(0, 0; z)H(0; y) + 2H(0, 0; z)H(1 - z, 0; y) \\ & + 4H(0, 0, 0; y)H(0; z) - 8H(0, 0, 0; z) + 4H(0, 0, 0; z)H(0; y) + 8H(0, 0, 0, 0; z) \\ & + 2H(0, 0, 1, 0; y) + 4H(0, 0, 1, 0; z) - 2H(0, 1, 0; y) - 4H(0, 1, 0; z) \\ & + 4H(0, 1, 0; z)H(0; y) + 2H(0, 1, 0; z)H(1 - z; y) + 2H(0, 1, 0, 0; y) + 4H(0, 1, 0, 0; z) \end{aligned}$$

Per arrivare a queste
formule
sono stati necessari
mesi di tempo
macchina

$$\begin{aligned}
& -H(0, 1, 1, 0; y) + 4H(0, 1, 1, 0; z) + 2H(0, 1 - z; y)H(1, 0; z) \\
& + 2H(0, 1 - z, 0; y)H(0; z) + 2H(0, 1 - z, 1, 0; y) + 4H(1, 0; z) - 2H(1, 0; z)H(1 - z; y) \\
& + 2H(1, 0; z)H(1 - z, 0; y) + H(1, 0; z)H(1 - z, 1 - z; y) - 4H(1, 0, 0; z) \\
& + 2H(1, 0, 0; z)H(1 - z; y) + 4H(1, 0, 0, 0; z) + 4H(1, 0, 1, 0; z) - 4H(1, 1, 0; z) \\
& + 2H(1, 1, 0; z)H(1 - z; y) + 4H(1, 1, 0, 0; z) + 4H(1, 1, 1, 0; z) + H(1 - z, 0, 1, 0; y) \\
& - 2H(1 - z, 1, 0; y) + 2H(1 - z, 1, 0, 0; y) - H(1 - z, 1, 1, 0; y) + H(1 - z, 1 - z, 1, 0; y) \\
& + \frac{7\pi^4}{90} + 5\zeta_3 H(0; z) \\
& + \frac{\pi^2}{6} \left[+ 4 - 4H(0; y) + 4H(0; y)H(0; z) + 4H(0; y)H(1; z) - 6H(0; z) \right. \\
& + H(0; z)H(1 - z; y) + 4H(0, 0; y) + 6H(0, 0; z) - H(0, 1; y) \\
& + 2H(0, 1 - z; y) - 4H(1; z) + 2H(1; z)H(1 - z; y) + 4H(1, \\
& \left. - 2H(1 - z; y) + 2H(1 - z, 0; y) - H(1 - z, 1; y) + H(1 - z
\end{aligned}$$

Ognuno degli $H(\dots)$ è
una funzione da
valutare
numericamente

$$\begin{aligned}
& -2H(1 - z, 0, 1, 0; y) + 2H(1 - z, 1, 0; y) - 2H(1 - z, 1, 0, 0; y) \\
& - H(1 - z, 1 - z, 1, 0; y) + \frac{37\pi^4}{360} + \zeta_3 [10 - 6H(0; y) - 5H(0; z) - H(1; z) - H(1 - z; y)] \\
& + \frac{\pi^2}{6} \left[+ 2H(0; y) - H(0; y)H(0; z) - H(0; y)H(1; z) - H(0; z)H(1 - z; y) \right. \\
& - 2H(0, 0; y) - H(0, 1; z) - 2H(0, 1 - z; y) + 2H(1; z) \\
& \left. - H(1, 0; z) + 2H(1 - z; y) - H(1 - z, 0; y) - H(1 - z, 1 - z; y) \right] .
\end{aligned} \tag{4.31}$$



$$= \left(\frac{S_\epsilon}{16\pi^2} \right)^2 \frac{(-s_{123})^{-2\epsilon}}{s_{23}} \sum_{i=-1}^3 \frac{f_{5.2,i} \left(\frac{s_{13}}{s_{123}}, \frac{s_{23}}{s_{123}} \right)}{\epsilon^i} + \mathcal{O}(\epsilon^2), \tag{4.26}$$

with:

$$f_{5.2,3}(y, z) = -1, \tag{4.27}$$

$$f_{5.2,2}(y, z) = -2 + H(0; y) + H(0; z), \tag{4.28}$$

$$f_{5.2,1}(y, z) = -4 + 2H(0; y) - H(0; y)H(0; z) + 2H(0; z) - 2H(0, 0; y) - H(0, 0; z) - H(1, 0; y), \tag{4.29}$$

$$\begin{aligned}
f_{5.2,0}(y, z) = & -8 + 4H(0; y) - 2H(0; y)H(0; z) + H(0; y)H(1, 0; z) + 4H(0; z) + H(0; z)H(1 - z, 0; y) \\
& - 4H(0, 0; y) + 2H(0, 0; y)H(0; z) - 2H(0, 0; z) + H(0, 0; z)H(0; y) + 4H(0, 0, 0; y) \\
& + H(0, 0, 0; z) + 2H(0, 1, 0; y) - 2H(1, 0; y) + H(1, 0; z)H(1 - z; y) + 2H(1, 0, 0; y) \\
& + H(1, 1, 0; z) + H(1 - z, 1, 0; y) + 5\zeta_3 + \frac{\pi^2}{6} [+ H(0; y) + H(1; z) + H(1 - z; y)] , \tag{4.30}
\end{aligned}$$

$$\begin{aligned}
f_{5.2,-1}(y, z) = & -16 + 8H(0; y) - 4H(0; y)H(0; z) + 2H(0; y)H(1, 0; z) - H(0; y)H(1, 0, 0; z) \\
& - H(0; y)H(1, 1, 0; z) + 8H(0; z) + 2H(0; z)H(1 - z, 0; y) - 2H(0; z)H(1 - z, 0, 0; y) \\
& - H(0; z)H(1 - z, 1 - z, 0; y) - 8H(0, 0; y) + 4H(0, 0; y)H(0; z) - 2H(0, 0; y)H(0, 0; z) \\
& - 2H(0, 0; y)H(1, 0; z) - 4H(0, 0; z) + 2H(0, 0; z)H(0; y) - H(0, 0; z)H(1 - z, 0; y) \\
& + 8H(0, 0, 0; y) - 4H(0, 0, 0; y)H(0; z) + 2H(0, 0, 0; z) - H(0, 0, 0; z)H(0; y) \\
& - 8H(0, 0, 0, 0; y) - H(0, 0, 0, 0; z) - 4H(0, 0, 1, 0; y) + 4H(0, 1, 0; y) \\
& - H(0, 1, 0; z)H(0; y) - H(0, 1, 0; z)H(1 - z; y) - 4H(0, 1, 0, 0; y) - H(0, 1, 1, 0; z) \\
& - 2H(0, 1 - z; y)H(1, 0; z) - 2H(0, 1 - z, 0; y)H(0; z) - 2H(0, 1 - z, 1, 0; y) \\
& - 4H(1, 0; y) + 2H(1, 0; z)H(1 - z; y) - H(1, 0; z)H(1 - z, 0; y) \\
& - H(1, 0; z)H(1 - z, 1 - z; y) + 4H(1, 0, 0; y) - H(1, 0, 0; z)H(1 - z; y) \\
& - 4H(1, 0, 0, 0; y) - H(1, 0, 1, 0; z) + 2H(1, 1, 0; z) - H(1, 1, 0, 0; z)
\end{aligned}$$

Risolvere numericamente un problema richiede
la capacità di spezzarlo in parti più semplici,
di individuare i passi per ottenere la risposta,
di controllare la correttezza di ogni passo

Non vi laureerete senza imparare a programmare

Cominciamo subito

<http://www.to.infn.it/~maina/>